

8

FURTHER APPLICATIONS OF INTEGRATION

fppt.com

FURTHER APPLICATIONS OF INTEGRATION

- **Here, we explore:**

  Some of the many other geometric applications of integration—such as the length of a curve and the area of a surface

fppt.com

8.1
Arc Length

In this section, we will learn about:

Arc length and its function.

fppt.com

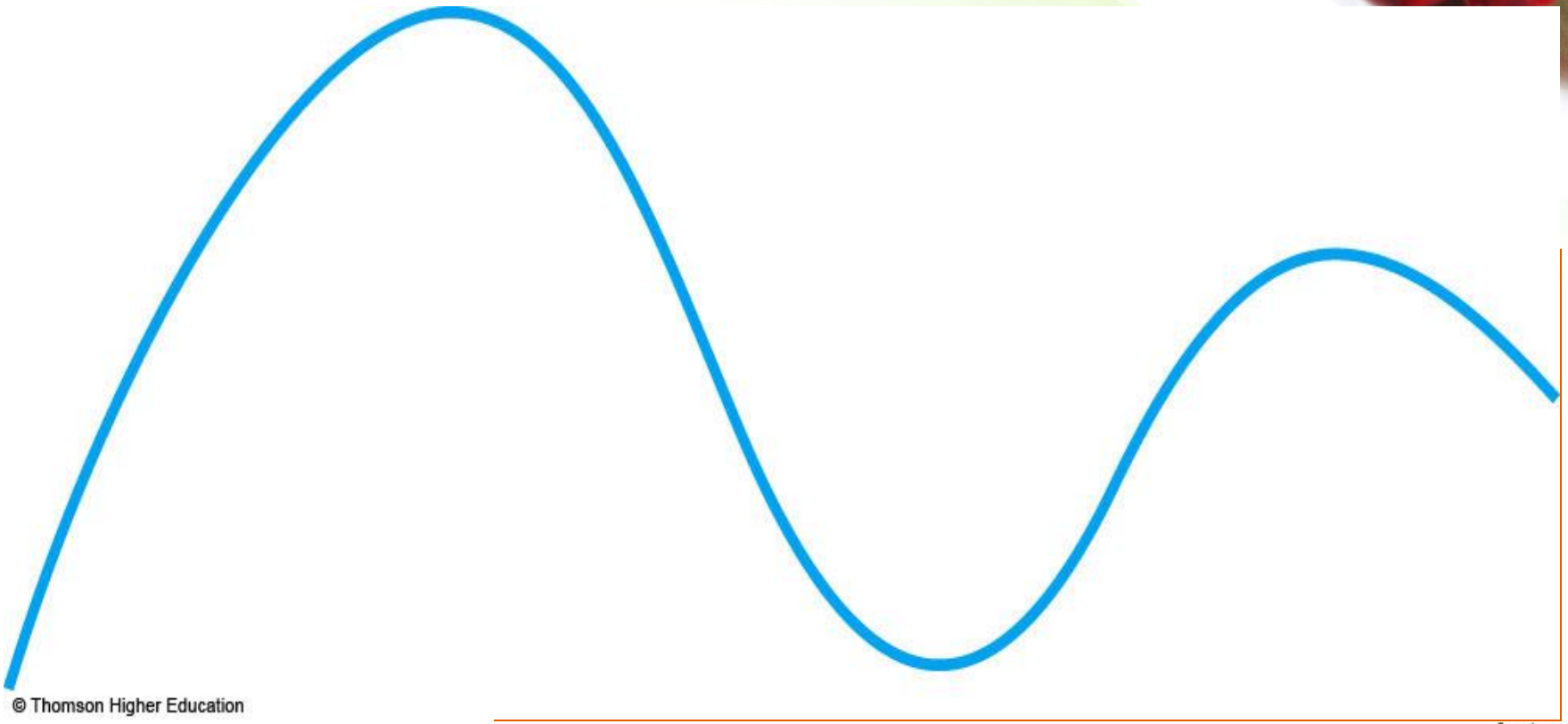
ARC LENGTH

- **What do we mean by the length of a curve?**

fppt.com

ARC LENGTH

- **We might think of fitting a piece of string to the curve here and then measuring the string against a ruler.**



© Thomson Higher Education

fppt.com

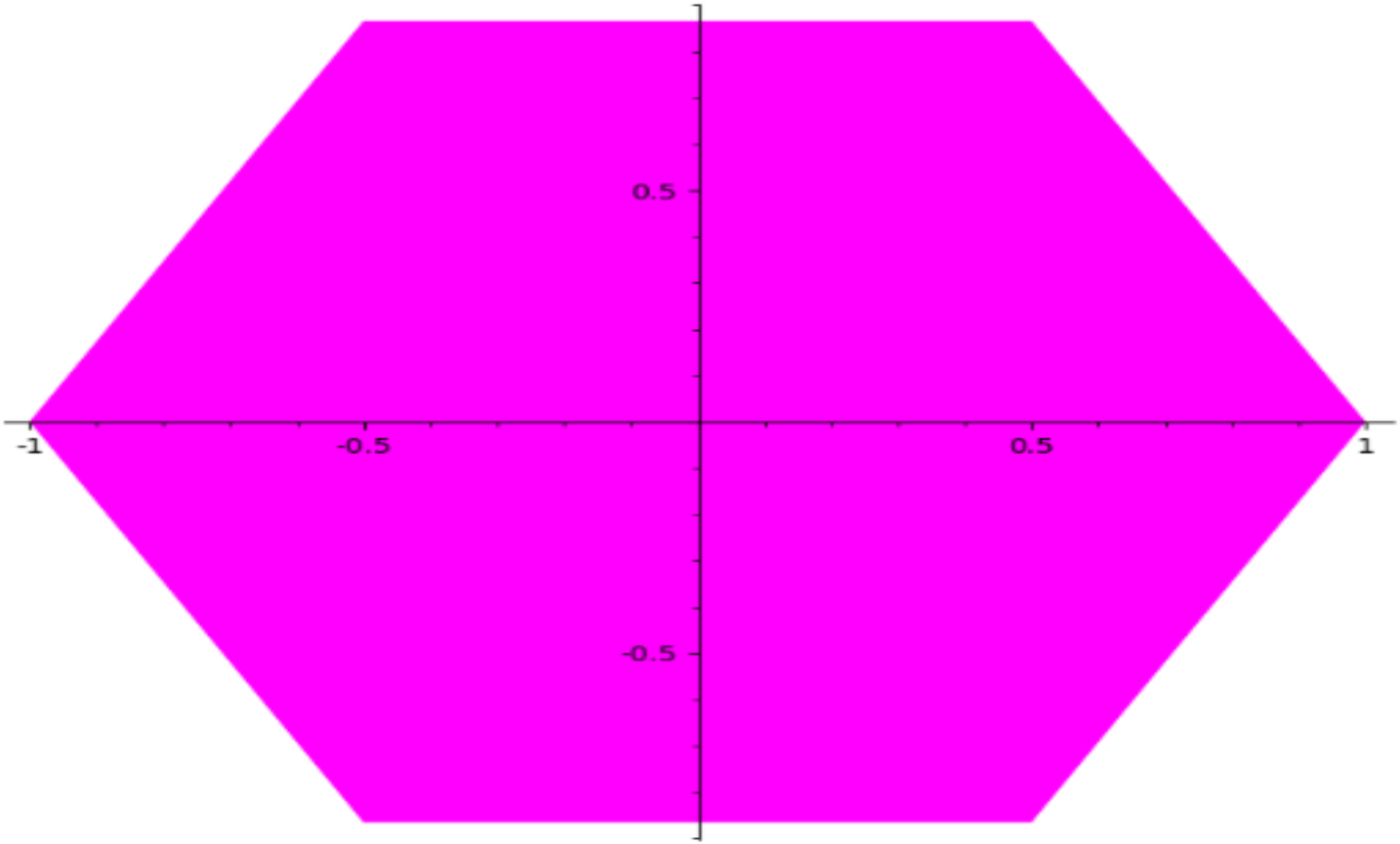
ARC LENGTH

- **However, that might be difficult to do with much accuracy if we have a complicated curve.**
- **We need a precise definition for the length of an arc of a curve—in the same spirit as the definitions we developed for the concepts of area and volume.**

fppt.com

POLYGON

- **If the curve is a polygon, we can easily find its length.**



ARC LENGTH FORMULA

Formula 2

- **If *f'* is continuous on [*a*, *b*], then the length of the curve *y* = *f*(*x*), *a* ≤ *x* ≤ *b* is:**

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx$$

fppt.com

ARC LENGTH FORMULA

Equation 3

- If we use Leibniz notation for derivatives, we can write the arc length formula as:

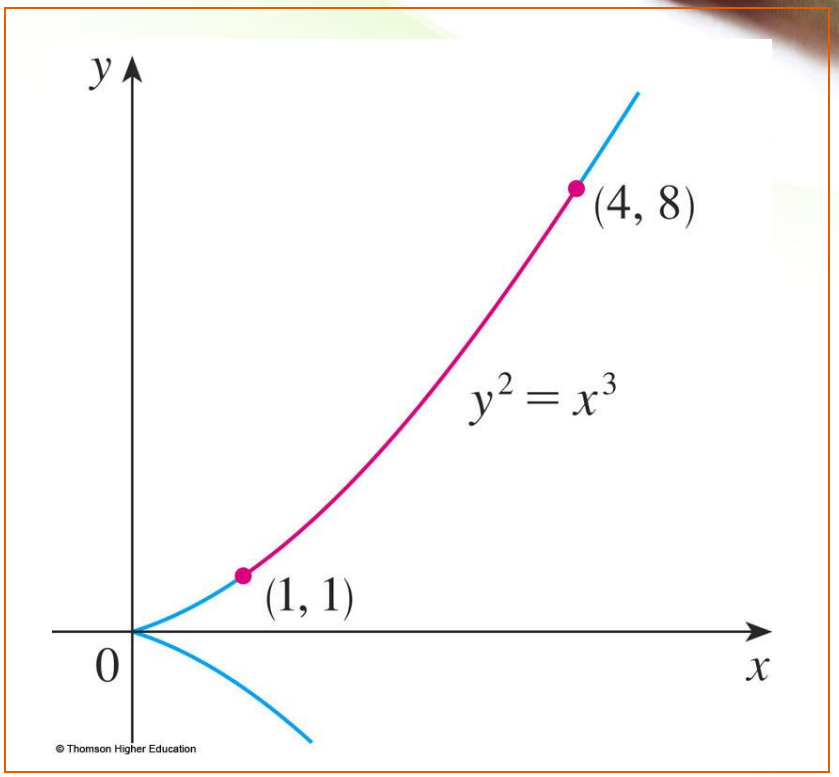
$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

fppt.com

Example 1

- **Find the length of the arc of the semi cubical parabola $y^2 = x^3$ between the points (1, 1) and (4, 8).**

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx$$



fppt.com

ARC LENGTH

Formula 4

- **If a curve has the equation $x = g(y)$, $c \le y \le d$, and $g'(y)$ is continuous, then by interchanging the roles of x and y in Formula 2 or Equation 3, we obtain its length as:**

$$L = \int_c^d \sqrt{1+[g'(y)]^2}\,dy = \int_c^d \sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$$

fppt.com

Example 2

- **Find the length of the arc of the parabola $y^2 = x$ from (0, 0) to (1, 1).**

fppt.com

Example

- Find

$$X=\frac{1}{3}(y^2+2)^{\frac{3}{2}} \qquad 0\leq y\leq 4$$

fppt.com

ARC LENGTH FUNCTION

- **We will find it useful to have a function that measures the arc length of a curve from a particular starting point to any other point on the curve.**

fppt.com

ARC LENGTH FUNCTION

- So, suppose a smooth curve C has the equation $y = f(x)$, $a \leq x \leq b$.

- Then, let $s(x)$ be the distance along C from the initial point $P_0(a, f(a))$ to the point $Q(x, f(x))$.

fppt.com

THE ARC LENGTH FUNCTION

- Then, *s* is a function, called the arc length function, and, by Formula 2,

Equation 5

$$s(x) = \int_a^x \sqrt{1 + [f'(t)]^2}\, dt$$

  - We have replaced the variable of integration by *t* so that *x* does not have two meanings.

fppt.com

ARC LENGTH FUNCTION

Equation 6

- We can use Part 1 of the Fundamental Theorem of Calculus (FTC 1) to differentiate Equation 5 (as the integrand is continuous):

$$\frac{ds}{dx} = \sqrt{1+\left[f'(x)\right]^2} = \sqrt{1+\left(\frac{dy}{dx}\right)^2}$$

fppt.com

ARC LENGTH FUNCTION

- Equation 6 shows that the rate of change of s with respect to x is always at least 1 and is equal to 1 when $f'(x)$, the slope of the curve, is 0.

fppt.com

ARC LENGTH FUNCTION

Equation 7

- The differential of arc length is:

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

fppt.com

ARC LENGTH FUNCTION

Equation 8

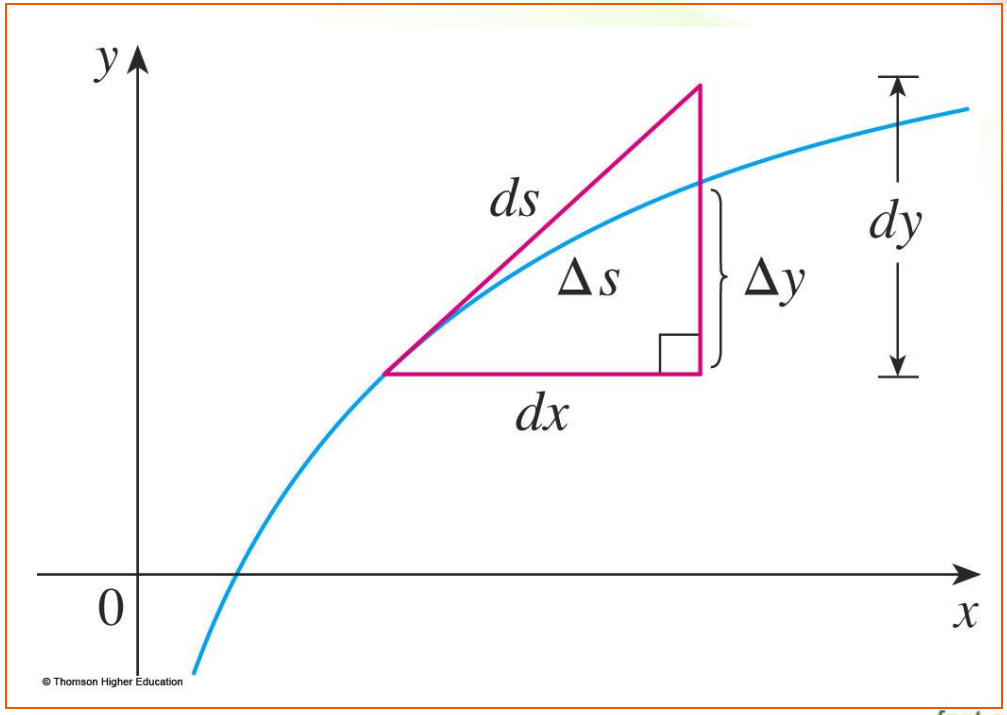
- Equation 7 is sometimes written in the symmetric form

$$(ds)^2 = (dx)^2 + (dy)^2$$

fppt.com

ARC LENGTH FUNCTION

- The geometric interpretation of Equation 8 is shown here.
  - It can be used as a mnemonic device for remembering both Formulas 3 and 4.



fppt.com

ARC LENGTH FUNCTION

- If we write $L = \int ds$, then, from Equation 8, we can either solve to get:
  - Equation 7, which gives Formula 3.
  - $ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$, which gives Formula 4.

fppt.com

ARC LENGTH FUNCTION

Example 4

- Find the arc length function for the curve $y = x^2 - \frac{1}{8} \ln x$ taking $P_0(1, 1)$ as the starting point.

fppt.com

ARC LENGTH FUNCTION

Example 4

- If $f'(x) = x^2 - \frac{1}{8}\ln x$, then

$$f'(x) = 2x - \frac{1}{8x}$$

$$1 + [f'(x)]^2 = 1 + \left(2x - \frac{1}{8x}\right)^2 = 1 + 4x^2 - \frac{1}{2} + \frac{1}{64x^2}$$

$$= 4x^2 + \frac{1}{2} + \frac{1}{64x^2}$$

$$= \left(2x + \frac{1}{8x}\right)^2$$

$$\sqrt{1 + [f'(x)]^2} = 2x + \frac{1}{8x}$$

fppt.com

ARC LENGTH FUNCTION

Example 4

- Thus, the arc length function is given by:

$$s(x) = \int_1^x \sqrt{1+\left[f'(t)\right]^2}\,dt$$

$$= \int_1^x \left(2t - \frac{1}{8t}\right)dt$$

$$= t^2 + \tfrac{1}{8}\ln t \Big]_1^x$$

$$= x^2 + \tfrac{1}{8}\ln x - 1$$

fppt.com

ARC LENGTH FUNCTION

Example 4

- For instance, the arc length along the curve from (1, 1) to (3, $f(3)$) is:

$$s(3) = 3^2 + \tfrac{1}{8}\ln 3 - 1$$

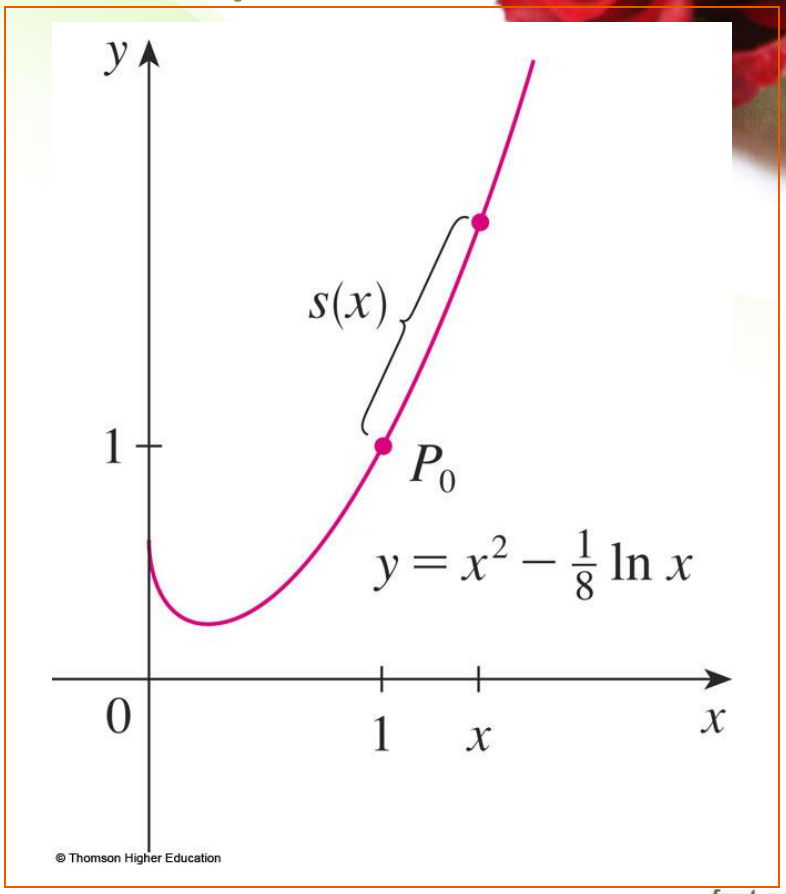
$$= 8 + \frac{\ln 3}{8}$$

$$\approx 8.1373$$

fppt.com

ARC LENGTH FUNCTION

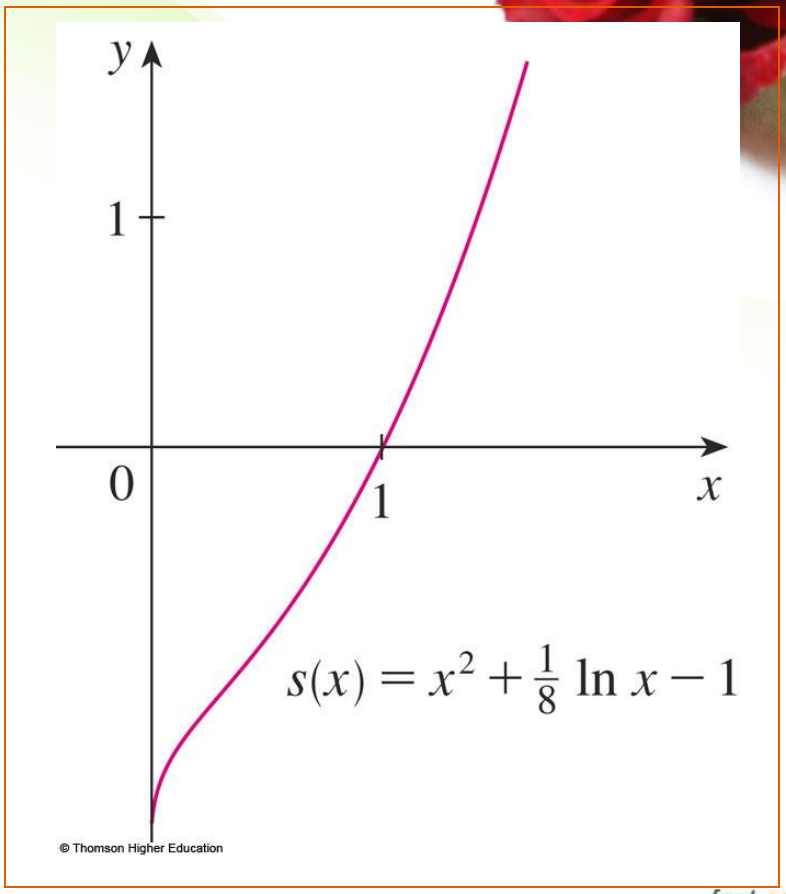
- The figure shows the interpretation of the arc length function in Example 4.



fppt.com

ARC LENGTH FUNCTION

- This figure shows the graph of this arc length function.
  - Why is $s(x)$ negative when x is less than 1?



fppt.com